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Subject: Programming with Java

ASSIGNMENT No : 4

TITLE : Wrapper Classes in Java

Problem Statement

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Learning Objectives

To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Programme

```

import java.util.Scanner;

public class WrapperDemo {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Input primitive values
        System.out.print("Enter an integer: ");
        int num = sc.nextInt();

        System.out.print("Enter a decimal number: ");
        double d = sc.nextDouble();

        // Primitive to Wrapper (Autoboxing)
        Integer intObj = num;
        Double doubleObj = d;

        // Display wrapper objects
        System.out.println("\nWrapper Objects:");
        System.out.println("Integer Object = " + intObj);
        System.out.println("Double Object = " + doubleObj);

        // Wrapper to Primitive (Unboxing)
        int intValue = intObj;
        double doubleValue = doubleObj;

        // Basic operations
        System.out.println("\nOperations:");
        System.out.println("Square of Integer = " + (intValue * intValue));
        System.out.println("Double Value + 10 = " + (doubleValue + 10));

        // Convert Wrapper to String
        String str = intObj.toString();
        System.out.println("\nInteger as String = " + str);

        // Convert String to Wrapper
        Integer newInt = Integer.valueOf(str);
        System.out.println("String converted back to Integer = " + newInt);

        sc.close();
    }
}

```

Output

```
PS C:\CS-JAVA\.vscode> javac WrapperDemo.java
PS C:\CS-JAVA\.vscode> java WrapperDemo
Enter an integer: 12
Enter a decimal number: 12.5

Wrapper Objects:
Integer Object = 12
Double Object = 12.5

Operations:
Square of Integer = 144
Double Value + 10 = 22.5

Integer as String = 12
String converted back to Integer = 12
PS C:\CS-JAVA\.vscode> 
```

Conclusion

The Java application successfully demonstrates the use of wrapper classes to convert primitive data types into objects and objects back into primitive values through boxing and unboxing. It also performs basic operations on wrapped values, showing how wrapper classes provide useful methods for handling primitive data types in object-oriented programming. This experiment helps in understanding the importance and applications of wrapper classes in Java.